(6,2): two triple blocks and six bands

5 recorded steps | 5 individual operations | 8 external adjacent checks

Let A be the full twist on123 and B the full twist on456. Let H be the ordered half twists14,25,36 and r=sigma5 sigma4 sigma3 sigma2 sigma1. The checked block-rotation identity is r^3=ABH. The SAME H exchanges A and B individually. Thus the six-point boundary twist is (ABH)^2, and the retained actual carries give A^2 B^2 H^2.

This yields two squared triple twists and six unit half twists, type(6,2). The full six-point disk boundary twist has identity marked-sphere action and chosen+I lift. The disk derivation and those closure checks are separate.

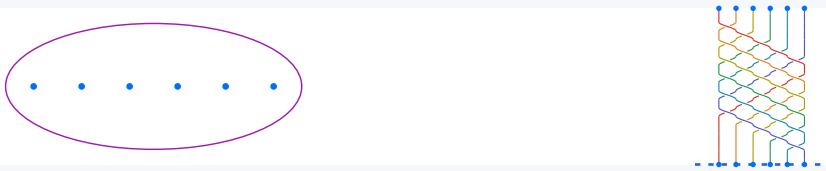
Reading the diagrams. Half-twist arcs have H labels; closed curves have T labels. Ordinary triple twists are temporary calculation factors. All factor grids have SIX UNNUMBERED points at18pt pitch and1.26pt radius. The connected braid represents the exact displayed factor words. Individual supports are not a joint cut-system or jointly minimal drawing.

Certificate scope. Faithful disk actions, raw free-word reconstructions, complete actual kernels AND shadows when the last strand is fixed, full actions for halves that move it, and separate sphere/lift checks remain in 6-2.json. Internal named-rule checks are retained separately from the external-check count. No unchosen framed-half-hole convention is asserted.

The explicit starting construction

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.

1. D6
T6



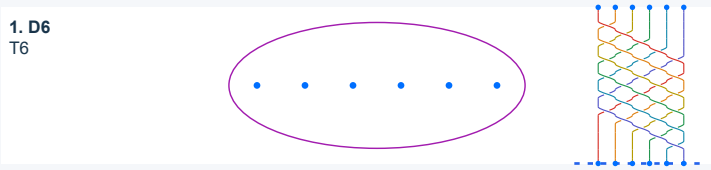
D - step 001

Use two half-turn packets: r^3 = A B H14 H25 H36

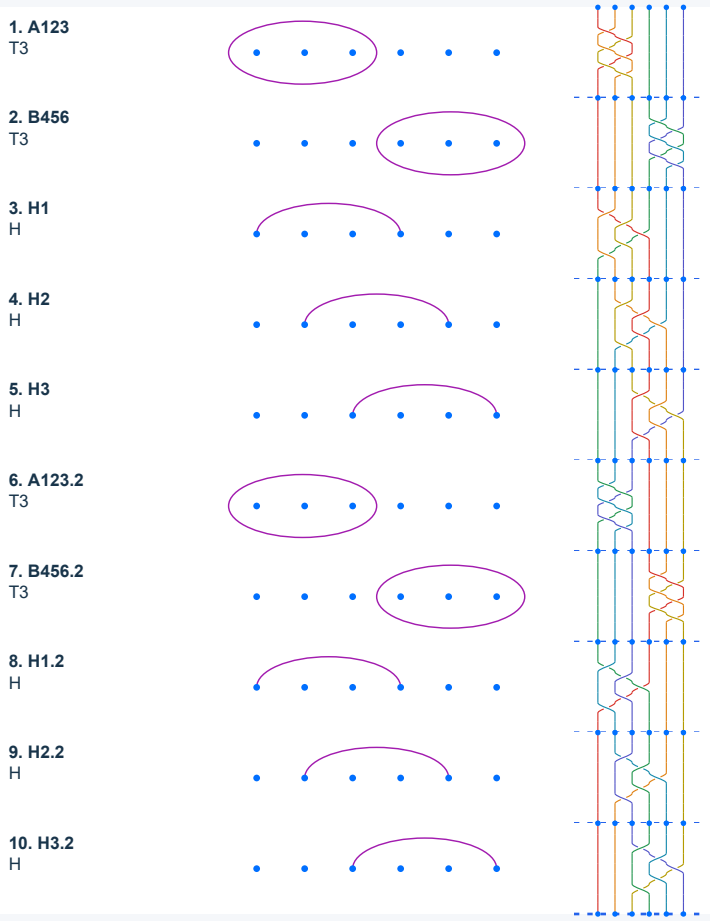
DISK SUBSTITUTION. Every ordered actual input/output and the full named local proof are checked. No sphere redraw or deletion is included.

0 unchanged displayed factors above; 0 below. Only the changed interval is shown.

Before



After



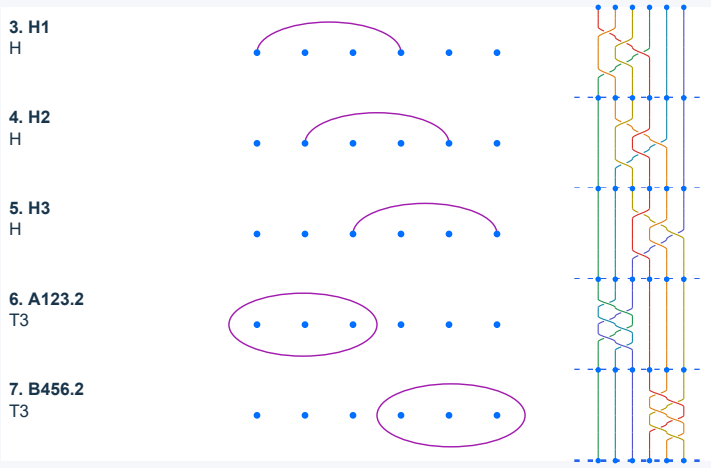
D - step 002

Carry the SAME three-half block down across both triple roots

DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

2 unchanged displayed factors above; 3 below. Only the changed interval is shown.

Before



After



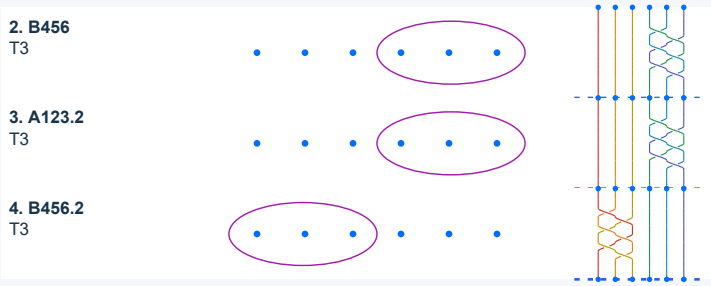
D - step 003

Commute the returned A root upward across the two disjoint B roots

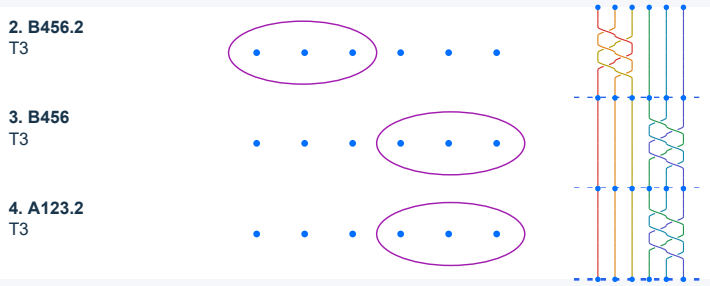
DISK TRANSPORT. Every adjacent crossing is raw-reconstructed and bounded20 before the complete carry is applied; all context changes remain recorded.

1 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



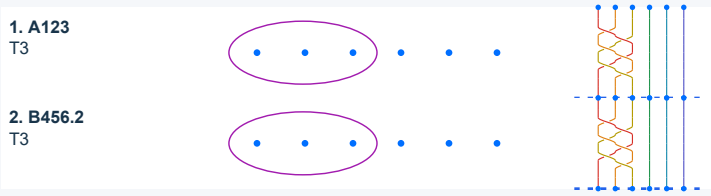
D - step 004

Combine the two matched A123.square

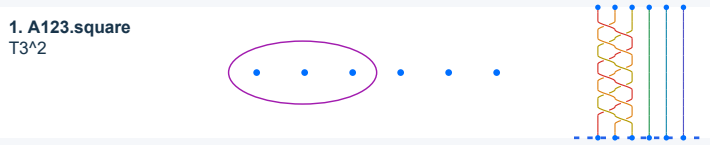
DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

0 unchanged displayed factors above; 8 below. Only the changed interval is shown.

Before



After



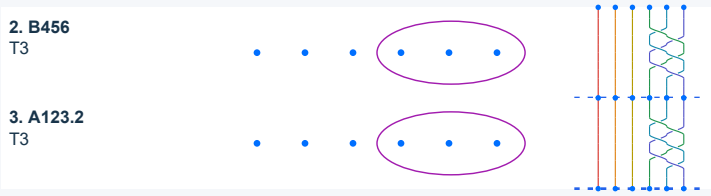
D - step 005

Combine the two matched B456.square

DISK COMBINE. The two supported twists are individually matched before their positive powers combine.

1 unchanged displayed factors above; 6 below. Only the changed interval is shown.

Before



After



Complete final type (6,2) factorization

FULL ORDERED FACTORIZATION. Read downward. Braids remain connected, with strand colors retained; disjoint crossings may share a level.



Closure certificates and precise scope

The final product has the exact marked-sphere identity action and the separately checked chosen genus-two lift +I. This is not a claim of equality to the trivial disk braid, an unquotiented spherical-braid equality, a framed section lift, or an unproved identification with a particular literature curve chart.

Final chosen lift matrix: [[1, 0, 0, 0], [0, 1, 0, 0], [0, 0, 1, 0], [0, 0, 0, 1]]

Source-pinned full certificate: 6-2.json; graphical viewer: 6-2.html. Imported source manifest: ../fibration-sources/planar-mcg-2026-10-30/provenance.json. All sources, exact actual input/output matches, canonical PC daisies/half lanterns and parallel-copy corrections remain available with the repository.